Exam Date & Time: 20-Feb-2026 (01:15 PM - 03:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computer Science and Applications
Master of Computer Applications (MCA)
Semester I - University Theory Examination (Supplementary Exam - 1) February 2026

DATABASE TECHNOLOGIES [CAUC508]

Marks: 50 **Duration: 135 mins.**

Section - I

Answer all the questions. Section Duration: 20 mins

1 Which of the following is NOT a section of a PL/SQL block?

1) Compilation 2) Declaration 3) Exception 4) Execution (1)

Course Outcome: "CO1" / Cognitive Level: "Understand"

2 Which of the following is NOT a conditional control?

1) IF 2) CASE 3) ELSIF 4) LOOP (1)

Course Outcome: "CO1" / Cognitive Level: "Application"

3 Which attribute returns TRUE if the cursor has no rows?

1) %ISOPEN 2) %NOTFOUND 3) %FOUND 4) %ROWTYPE (1)

Course Outcome: "CO2" / Cognitive Level: "Application"

4 Triggers are executed _____.

1) Automatically 2) Only once per session 3) Only on demand 4) Manually (1)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5 A graph database represents data using_____

1) Rows 2) Flat files 3) Edges and nodes 4) Columns (1)

Course Outcome: "CO3" / Cognitive Level: "Understand"

6 Which command is used to remove a relation from an SQL

1) Delete 2) Remove 3) Update 4) Drop (1)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

7 In MongoDB, the term "schema-less" means_____ (1)

1) Each document can have a dynamic structure 2) All documents must have identical fields 3) Documents cannot have fields 4) No need for indexes

Course Outcome: "CO4" / Cognitive Level: "Application"

8

Which of the following is true about %TYPE?

1)	Declares a variable without size		2)	Declares an exception		3)	Used for loops only		4)	Declares a variable with the same datatype as a database column	

(1)

Course Outcome: "CO4" / Cognitive Level: "Understand"

9

What is TRUE about %ISOPEN for an implicit cursor?

1)	It is always TRUE		2)	It returns TRUE after COMMIT		3)	It throws an error if used		4)	It is always FALSE except during execution	

(1)

Course Outcome: "CO5" / Cognitive Level: "Application"

10

What does db.collection.findOne() return?

1)	Cursor		2)	Array of documents		3)	Only the first document		4)	Last document	

(1)

Course Outcome: "CO5" / Cognitive Level: "Understand"

Section - II

Answer 4 out of 6 questions.

1

Apply any five Codd's rules to evaluate whether a given DBMSfollows the relational model.

(5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

2

Explain the CAP theorem and describe its three components:Consistency, Availability, and Partition Tolerance.

(5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

3

Analyze how replication improves availability and fault tolerancein distributed databases.

(5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4

Evaluate HBase as a storage solution for big data applicationscompared to relational databases

(5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

5

Evaluate the role of ACID properties in maintaining data integrityin a transactional database system.

(5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

6

Analyze how Cassandra achieves high availability and faulttolerance using its peer-to-peer architecture.

(5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

Section - III

Answer 4 out of 6 questions.

1		

Difference between MongoDB and RDBMS

(5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

2		

Explain the Neo4j data model and its architecture.

(5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

3		

Design NOSQL database for social media.

(5)

Course Outcome: "CO3" / Cognitive Level: "Create"

4		

Design a PL/SQL program using nested loops to generate amultiplication table.

(5)

Course Outcome: "CO4" / Cognitive Level: "Create"

5		

Analyze how the three-level architecture supports data independencein DBMS.

(5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

6		

Assess the roles of different database actors

(5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
University Exam Dec-Feb 2026	CAUC508:DATABASE TECHNOLOGIES

#	Course Outcome	Description	Marks
1	CO1	Understand basic concepts regarding data, database systems and various data models.	12
2	CO2	Enhance the knowledge of the processes of Database Development and Administration using SQL and PL/SQL	12
3	CO3	Able to understand different types of NoSQL Databases	12
4	CO4	Architectures and common features of the basic types of NoSQL databases	12
5	CO5	Able to design Schema and implement CRUD operations, distributed data operations, and implement various column store internals	22
		Total	**70**

* NA - CO Not Selected

Event Name	Subject
University Exam Dec-Feb 2026	CAUC508:DATABASE TECHNOLOGIES

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	7	19(27.14%)
3	Application	5	9(12.86%)
4	Analysis	5	21(30%)

5	Create	2	10(14.29%)
6	Evaluate	3	11(15.71%)
	Total	22	70 (100%)